



Fig. 6: Flow-chart of the program

Battery voltage. With battery disconnected, apply 20V at TP2 with respect to TP0. Use a multimeter to monitor the voltage at test point TP3 and adjust preset VR1 to get 5V. Check whether the voltage at TP6 is around 5V. Connect back the batteries. Now the voltage at TP3 should be around 3V.

Solar voltage. Remove the solar panel from the circuit. Connect a 12V battery at test point TP4 with respect to TP0 and monitor the voltage at test point TP5. Adjust preset VR2 to get

3V. Relay (RL1) enable can be checked at TP7.

The circuit is ready to harness the energy of the sun after calibrations. The power is calculated and displayed on the LCD in watts every second. The energy in watt-second is calculated by integrating the power.

Fig. 5 shows the working prototype of PIC16F877-based solar charger.

The following parameters are cyclically displayed on the LCD module:

1. Battery voltage in millivolts
2. Battery current in milliamperes
3. Energy in watt-seconds
4. Power in watts
5. Solar panel voltage in millivolts
6. Charger mode: boost or trickle

Software

The source program is written in Basic language and compiled using PIC Simulator IDE from Oshonsoft. The IDE provides a facility to program using Basic like commands, then compile the program and generate hex code. Burn the generated hex code into the microcontroller by using a suitable programmer.

The program works as per the flow-chart shown in Fig. 6. It starts by checking solar panel voltage. If solar panel voltage is more than 12.6 volts, the program moves to the next stage. If solar panel voltage is less than 12.6 volts, the program displays message

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

"Low Solar Volts" on the LCD module and loops back to wait until solar panel voltage is above 12.6 volts.

If solar panel voltage is adequate, the system checks battery voltage and sets the charging mode to 'boost' or 'trickle.' A battery voltage of over 12V sets charging mode to 'trickle,' while a battery voltage of less than 12V sets it to 'boost' mode. During initialisation, the data is also read from the EEPROM, which stores the watt-hour readings. It gives an indication of the power absorbed from the sun.

The timer generates an interrupt every 65.56 ms. A 15-count in the interrupt service routine ensures that the energy and power are calculated every $65.56 \times 15 = 983.4$ ms (almost 1 second). The power is integrated every second to get the energy in watt-seconds. The watt-hour readings are stored in the EEPROM of the microcontroller, so that the data is not lost due to power failure. To prevent too many write cycles in the EEPROM, the data is stored only every 30 minutes.

The author is working at Tata Realty and Infrastructure in Engineering Services as AGM

SOLAR.BAS

```

*****
Solar.bas
*****
'RA2=SOLAR CELL VOLTAGE
'RA1=BATTERY CHARGE CURRENT IB
'RB0-RB7:LCD DATA LINES
'RC0=PWM OUTPUT FOR CHARGER
'RD1:LCD R/S
'RD2:LCD R/W
'RD3:LCD-E
'RD4:SPARE
'RD5:SPARE
'VARIABLE DECLARATION FOR LCD MODULE
Define LCD_BITS = 8
Define LCD_DREG = PORTB
Define LCD_DBIT = 0
Define LCD_RSREG = PORTD
Define LCD_RSBIT = 1
Define LCD_EREG = PORTD

Define LCD_EBIT = 3
Define LCD_RWREG = PORTD
Define LCD_RWBIT = 2
Define ADC_SAMPLEUS = 100
  
```

```

Define LCD_LINES = 2
Define LCD_CHARS = 16
'for simulation only
Define LCD_COMMANDUS = 5000 'delay after
LCDCMDOUT, default value is 5000
Define LCD_DATAUS = 100 'delay after
LCDOUT, default value is 100
Define LCD_INITMS = 20
'SETTING THE TRISA,TRISB AND ADCON1
REGISTERS
'ALL A PINS ARE INPUTS AND 3 E PORTS
AS INPUTS
TRISA = %00110111
ADCON1 = %00000000
TRISE = %00000111
ADCON0 = 0x80
TRISD = %11000000
TRISB = %00000000
INTCON = %10100000
OPTION_REG = %00000111
'VARIABLE DECLARATION
'FOR BATTERY VOLTAGE MEASUREMENT
Dim batt_curr As Word
Dim batt_curr_count As Byte
  
```

```

Symbol chg_on = PORTA.3
Dim boost_on As Bit
Dim trickle_on As Bit
Dim was_boost As Bit
Dim was_trickle As Bit
Dim trickle_time_count As Word
'FOR BATTERY AND SOLAR VOLTAGE
Dim batt_volt_raw As Word
Dim batt_volt As Word
Dim solar_volt_raw As Word
Dim solar_volt As Word
'FOR BATTERY CURRENT
Dim batt_curr_raw As Word
Dim batt_curr As Word
Dim batt_curr_data(5) As Word
Dim set_curr As Word
'SOLAR CELL MINIMUM VALUES
Dim batt_min_volt As Word
Dim solar_min_volt As Word
'POWER AND ENERGY CALCULATION
Dim power As Word
Dim energy As Word
Dim energy_aux As Word
Dim power_act As Word
  
```